

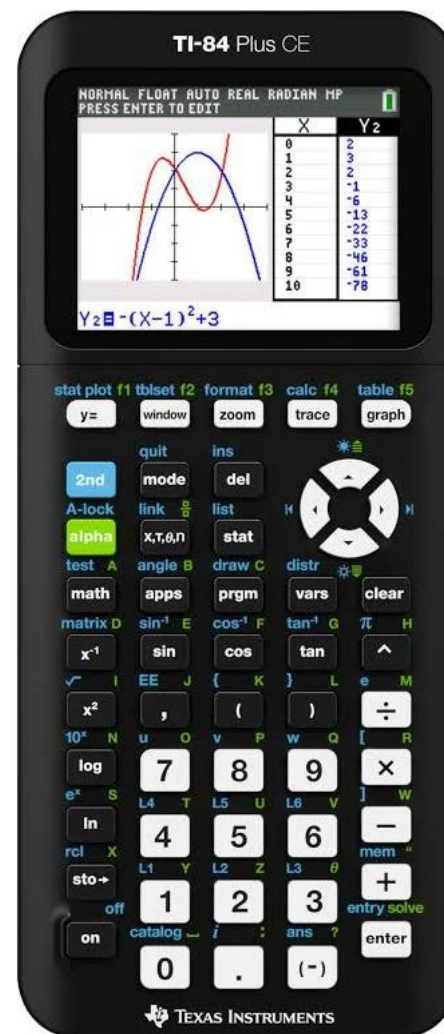


11 – Confidence interval & H0 test

Tutor: Chin Yi

Graphing Calculator

- If you have a **GC**, please use it for statistics questions (ie. From this tutorial onwards).
- Practice using it.
- ~~Beg, borrow, steal.~~



Recap: Normal dist vs t-dist

Theorem 1 (Confidence Interval for μ of normal distribution with known σ^2)

Let $X_1, \dots, X_n$ i.i.d $\sim N(\mu, \sigma^2)$. The $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Recall z_α is the upper $100(1 - \alpha)\%$ point of the standard normal, i.e. $\mathbb{P}(\phi > z_\alpha) = \alpha$.

Theorem 3 (Confidence Interval for μ of normal distribution with unknown σ^2 , $n \geq 30$)

Let $X_1, \dots, X_n$ i.i.d $\sim N(\mu, \sigma^2)$, $n \geq 30$, and σ^2 is *unknown*. The $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}},$$

where S^2 is the sample variance.

Here, we can use Normal Dist

Normally only when $n \geq 30$.

Recap: Normal dist vs t-dist

Theorem 4 (Confidence Interval for μ of normal distribution with unknown σ^2 , $n < 30$)

Let $X_1, \dots, X_n$ i.i.d $\sim N(\mu, \sigma^2)$, $n < 30$, and σ^2 is *unknown*. The $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}},$$

where S^2 is the sample variance.

We will use **n-1** degree of freedom when using the t – dist.

Recap: Summary

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

This will be on your finals cheat sheet.

Problem 1 (Confidence Intervals for Normal Distribution)

Suppose the fat content of certain steaks follows a $N(\mu, \sigma^2)$ distribution. The following observations $x_1, \dots, x_{16}$ for the fat content are given.

5.33, 4.25, 3.15, 3.70, 1.61, 6.39, 3.12, 6.59, 3.53, 4.74, 0.11, 1.60, 5.49, 1.72, 4.15, 2.28

- (i) Suppose $\sigma^2 = 3.2$. Find 90%, 95%, and 99% confidence intervals for μ based on the observations above.
- (ii) In part (i), if we want cut down the length of the confidence intervals to half their length, how much would we need to increase sample size?

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S/\sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S/\sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

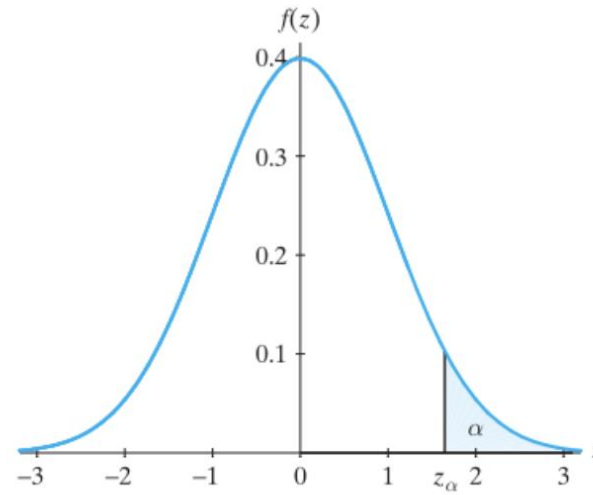
$$\bar{X} = \frac{1}{n} \sum x_i$$

For the given sample, we have $n = 16$, $\bar{X} = \underline{3.61}$, $\sigma = \sqrt{3.2}$.

1. For 90% confidence interval, $\alpha = 0.1$ and $z_{\alpha/2} = z_{0.05} \approx 1.645$. By (1), a 90% confidence interval for μ is

$$[2.874, 4.346].$$

Table Vb The Standard Normal Right-Tail Probabilities



$$P(Z > z_\alpha) = \alpha$$

$$P(Z > z) = 1 - \Phi(z) = \Phi(-z)$$

z_α	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367

Problem 2 (Confidence Intervals)

A random sample of 100 automobile owners in Tekong state shows that an automobile is driven on average 23,500 km per year with a standard deviation of 3900 km. Assume the population distribution is normal.

Construct a 99% confidence interval for the average number of km per year an automobile is driven in this state.

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

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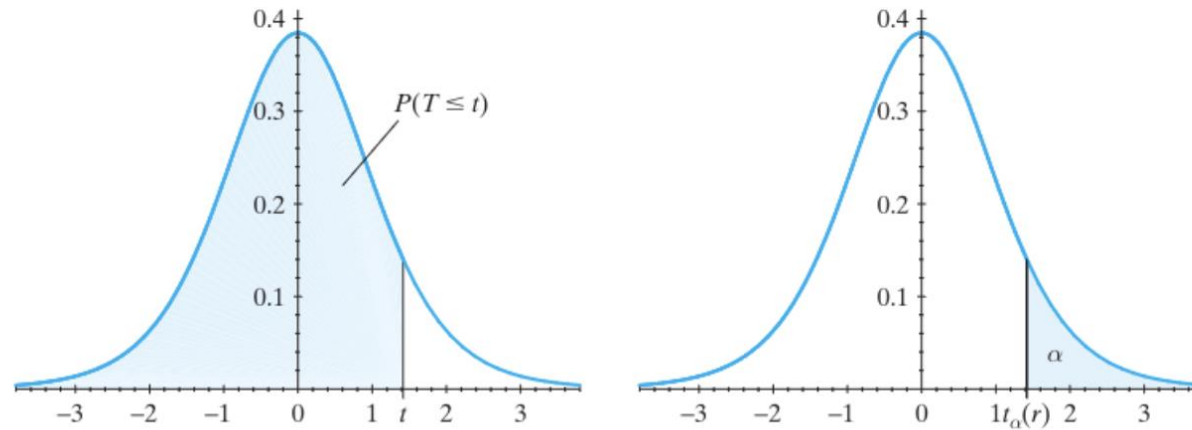
$$\begin{aligned}\bar{x} - z_{0.005} \frac{s}{\sqrt{n}} &< \mu < \bar{x} + z_{0.005} \frac{s}{\sqrt{n}} \\ 23500 - 2.575 \cdot \frac{3900}{\sqrt{100}} &< \mu < 23500 + 2.575 \cdot \frac{3900}{\sqrt{100}} \\ 22495.75 &< \mu < 24504.25\end{aligned}$$

Problem 3 (Confidence Intervals)

A random sample of 10 chocolate energy bar of a certain brand has, on average, 230 calories per bar, with a standard deviation of 15 calories. Construct a 99% confidence interval for the true mean calorie content of this brand of energy bar. Assume that the population distribution of the calorie content is normal.

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

Table VI The t Distribution



$$P(T \leq t) = \int_{-\infty}^t \frac{\Gamma[(r+1)/2]}{\sqrt{\pi r} \Gamma(r/2) (1 + w^2/r)^{(r+1)/2}} dw$$

$$P(T \leq -t) = 1 - P(T \leq t)$$

	$P(T \leq t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169

Problem 3 (Confidence Intervals)

A random sample of 10 chocolate energy bar of a certain brand has, on average, 230 calories per bar, with a standard deviation of 15 calories. Construct a 99% confidence interval for the true mean calorie content of this brand of energy bar. Assume that the population distribution of the calorie content is normal.

$$\bar{x} - t_{0.005}(9) \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{0.005}(9) \frac{s}{\sqrt{n}}$$

$$230 - 3.25 \cdot \frac{15}{\sqrt{10}} < \mu < 230 + 3.25 \cdot \frac{15}{\sqrt{10}}$$

$$214.58 < \mu < 245.42.$$

Problem 4 (Confidence Intervals)

Suppose the fat content of certain steaks follows a $N(\mu, \sigma^2)$ distribution. The following observations $x_1, \dots, x_{16}$ for the fat content are given.

5.33, 4.25, 3.15, 3.70, 1.61, 6.39, 3.12, 6.59, 3.53, 4.74, 0.11, 1.60, 5.49, 1.72, 4.15, 2.28

Suppose that both μ and σ^2 are *unknown*.

- (i) Find 90%, 95%, and 99% confidence intervals for μ .
- (ii) Find 90%, and 95% confidence intervals for σ^2 .

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

Problem 4 (Confidence Intervals)

Suppose the fat content of certain steaks follows a $N(\mu, \sigma^2)$ distribution. The following observations $x_1, \dots, x_{16}$ for the fat content are given.

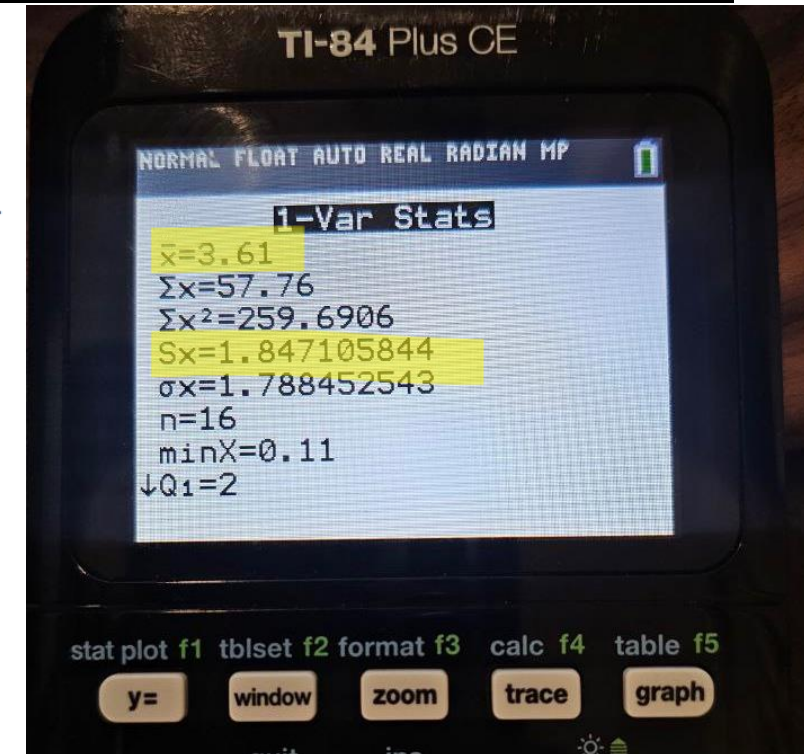
5.33, 4.25, 3.15, 3.70, 1.61, 6.39, 3.12, 6.59, 3.53, 4.74, 0.11, 1.60, 5.49, 1.72, 4.15, 2.28

Suppose that both μ and σ^2 are *unknown*.

- (i) Find 90%, 95%, and 99% confidence intervals for μ .
- (ii) Find 90%, and 95% confidence intervals for σ^2 .

$$\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \quad (2)$$

1. For 90% confidence interval, $\alpha = 0.1$ and $t_{\alpha/2}(n-1) \approx 1.753$. By (2), a 90% confidence interval for μ is
 $[2.800, 4.419]$.
2. For 95% confidence interval, $\alpha = 0.05$ and $t_{\alpha/2}(n-1) \approx 2.131$. By (2), a 95% confidence interval for μ is
 $[2.626, 4.594]$.
3. For 99% confidence interval, $\alpha = 0.01$ and $t_{\alpha/2}(n-1) \approx 2.947$. By (2), a 99% confidence interval for μ is
 $[2.249, 4.971]$.



$$S^2 = \frac{1}{n-1} \sum (\bar{x} - x_i)^2$$

Recap: Estimating sigma

Theorem 6 (Confidence Interval for σ^2 of normal distribution)

Let $X_1, \dots, X_n$ i.i.d $\sim N(\mu, \sigma^2)$. The $100(1 - \alpha)\%$ confidence interval for σ^2 is given by

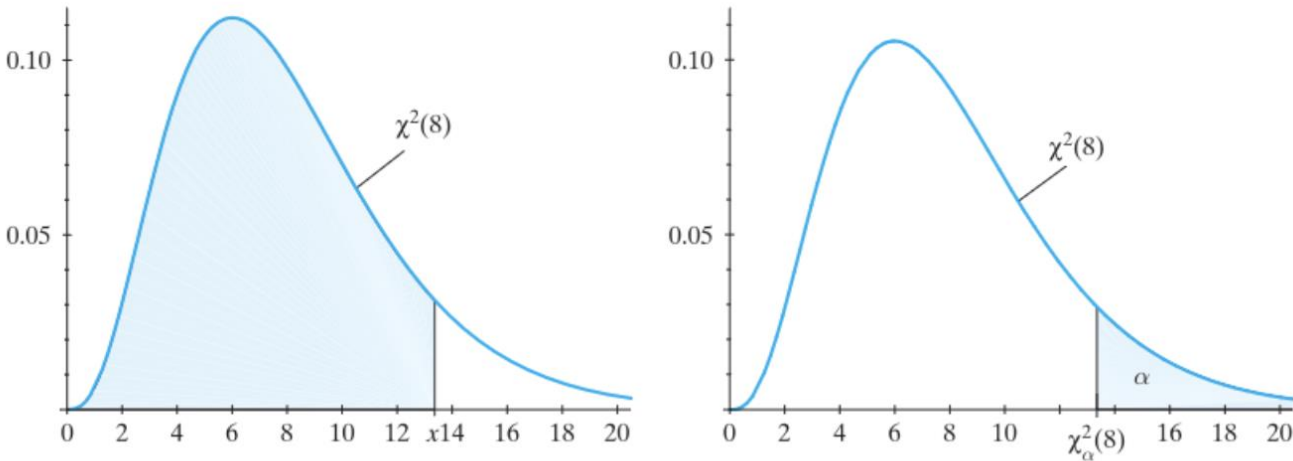
$$\frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)} < \sigma^2 < \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)},$$

where S^2 is the sample variance.

We **can't** use GC to find Confidence interval of σ^2 unfortunately.

We will use chi-square dist with **n-1** degree of freedom.

Table IV The Chi-Square Distribution



$$P(X \leq x) = \int_0^x \frac{1}{\Gamma(r/2)2^{r/2}} w^{r/2-1} e^{-w/2} dw$$

	$P(X \leq x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58

(ii) We have

$$X = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

For each $\alpha \in (0, 1)$, let $\chi_\alpha^2(n-1)$ denote the upper percentage point. Recall that the $100(1-\alpha)$ confidence interval is given by

$$\frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)} \leq \sigma_0^2 \leq \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)}. \quad (3)$$

Note that $n = 16$ and $S^2 \approx 3.412$.

1. For 90% confidence interval, we have $\alpha = 0.1$ and

$$\chi_{1-\alpha/2}^2(n-1) = \chi_{0.95}^2(15) \approx 7.261 \text{ and } \chi_{\alpha/2}^2(n-1) = \chi_{0.05}^2(15) \approx 25.$$

By (3), a 90% confidence interval for σ^2 is

$$[2.047, 7.050].$$

2. For 95% confidence interval, we have $\alpha = 0.05$ and

$$\chi_{1-\alpha/2}^2(n-1) = \chi_{0.975}^2(15) \approx 6.262 \text{ and } \chi_{\alpha/2}^2(n-1) = \chi_{0.025}^2(15) \approx 27.49.$$

By (3), a 95% confidence interval for σ^2 is

$$[1.861, 8.172].$$

Recap: Hypothesis Testing

- “at least as extreme as t ” (in the direction of rejection of H_0) means
 - $T \geq t$ (one-sided test)
 - $T \leq t$ (one-sided test)
 - $|T - \mathbb{E}[T]| \geq |t - \mathbb{E}[T]|$ (two-sided test)

One-sided Test

Compute p -value based on data:

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.02$$

$$\begin{aligned} p\text{-value} &= \mathbb{P}(T \geq t) \\ &= \mathbb{P}(T \geq 2.02) \\ &= 1 - \Phi(2.02) = 1 - 0.9783 = 0.0217 < \alpha. \end{aligned}$$

Decision: Reject H_0 (since the p -value is less than $\alpha = 0.05$).

Two-sided Test

Compute p -value based on data:

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{7.8 - 8}{0.5/\sqrt{50}} = -2.83$$

$$\begin{aligned} p\text{-value} &= \mathbb{P}(|T| \geq |t|) \\ &= \mathbb{P}(|T| \geq 2.83) \\ &= 2(1 - \Phi(2.83)) = 2 \times 0.0023 \approx 0.0046 < \alpha = 0.01. \end{aligned}$$

Decision: Reject H_0 (since the p -value is less than $\alpha = 0.01$).



Problem 5 (Hypothesis Testing)

In the journal Hypertension, researchers report that individuals who practice Transcendental Meditation (TM) lower their blood pressure significantly. If a random sample of 225 male TM practitioners meditate for 8.5 hours per week with a standard deviation of 2.5 hours, does that suggest that, on average, men who use TM meditate more than 8 hours per week? Use a significance level of $\alpha = 0.05$. State the null hypothesis, alternative hypothesis, test statistic and the conclusion.

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

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$$t = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

- p -value: (assuming H_0 is true)

$$\begin{aligned}\mathbb{P}(T \geq t) &= \mathbb{P}(T \geq 3.0) \\ &= 1 - \Phi(3) = 0.0013\end{aligned}$$

- Conclusion: Since the p -value is less than $\alpha = 0.05$, we reject the null hypothesis.

Problem 6 (Hypothesis Testing)

An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. A random sample of 30 bulbs has an average life of 788 hours. Use a 0.01 level of significance to test the hypothesis that $\mu = 800$ hours against the alternative hypothesis, $\mu \neq 800$ hours.

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

Problem 6 (Hypothesis Testing)

An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. A random sample of 30 bulbs has an average life of 788 hours. Use a 0.01 level of significance to test the hypothesis that $\mu = 800$ hours against the alternative hypothesis, $\mu \neq 800$ hours.

$$t = \frac{788 - 800}{40/\sqrt{30}} = -1.64.$$

$$t = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- p -value: (assuming H_0 is true)

$$\begin{aligned}\mathbb{P}(|T - \mathbb{E}[T]| \geq |t - \mathbb{E}[T]|) &= \mathbb{P}(|T - 0| \geq |-1.64 - 0|) \\ &= \mathbb{P}(|T| \geq 1.64) = \mathbb{P}(T \geq 1.64) + \mathbb{P}(T \leq -1.64) \\ &= 2 \times \mathbb{P}(T \leq -1.64) = 2\Phi(-1.64) = 2(0.0505) = 0.101 > \alpha = 0.05\end{aligned}$$

- Conclusion: Since the p -value is more than $\alpha = 0.05$, we **do not** reject the null hypothesis. There is no strong evidence to conclude that the mean lifetime is not 800.

Problem 7 (Hypothesis Testing)

Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a mean of 35 minutes. A random sample of 20 high school seniors took an average of 33.1 minutes to complete this test with a standard deviation of 4.3 minutes. Test the hypothesis that $\mu = 35$ against the alternative hypothesis that $\mu < 35$, at the 0.05 level of significance.

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Case	Statistic Q	Dist. of Q	CI
Known σ^2	$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$	$N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$
Unknown σ^2 ($n \geq 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	$\approx N(0, 1)$	$\left[\bar{X} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
Unknown σ^2 ($n < 30$)	$\frac{\bar{X} - \mu}{S / \sqrt{n}}$	t -distribution degree freedom $n-1$	$\left[\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}} \right]$

Problem 7 (Hypothesis Testing)

Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a mean of 35 minutes. A random sample of 20 high school seniors took an average of 33.1 minutes to complete this test with a standard deviation of 4.3 minutes. Test the hypothesis that $\mu = 35$ against the alternative hypothesis that $\mu < 35$, at the 0.05 level of significance.

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

- p -value: (assuming H_0 is true)

$$\begin{aligned}\mathbb{P}(T \leq t) &= \mathbb{P}(T \leq -1.976) \\ &< \mathbb{P}(T \leq -1.729) = \mathbb{P}(T \geq 1.729) = 0.05 = \alpha\end{aligned}$$

- Conclusion: Since the p -value is less than $\alpha = 0.05$, we **reject** the null hypothesis. There is strong evidence to conclude that the mean time required by high school senior to complete the standardized test is less than 35 minutes.

t-distribution Table

	$P(T \leq t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012
14	0.258	0.692	1.345	1.761	2.145	2.624	2.997
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845

Bonus Qn

$\mathbb{P}(H_0 \text{ rejected} | H_0)$ is also called the **size** of the test.

Problem 11.2

Let X_1, X_2 be i.i.d. $U(\theta, \theta + 1)$. The CDF of $Y = X_1 + X_2$ is,

$$F_Y(y) = \begin{cases} 0, & \text{if } y < 2\theta \\ \frac{1}{2}(y - 2\theta)^2, & \text{if } 2\theta \leq y < 2\theta + 1 \\ 1 - \frac{1}{2}(2\theta + 2 - y)^2, & \text{if } 2\theta + 1 \leq y < 2\theta + 2 \\ 1, & \text{if } y \geq 2\theta + 2 \end{cases}$$

For testing $H_0: \theta = 0$ against $H_1: \theta > 0$, we have two competing tests:

$\phi_1(X_1)$: Reject H_0 if $X_1 > 0.92$

$\phi_2(X_1, X_2)$: Reject H_0 if $X_1 + X_2 > C$

a) Find the value of C so that ϕ_2 has the same size as ϕ_1 .